

Deep Learning

Ali Ghodsi

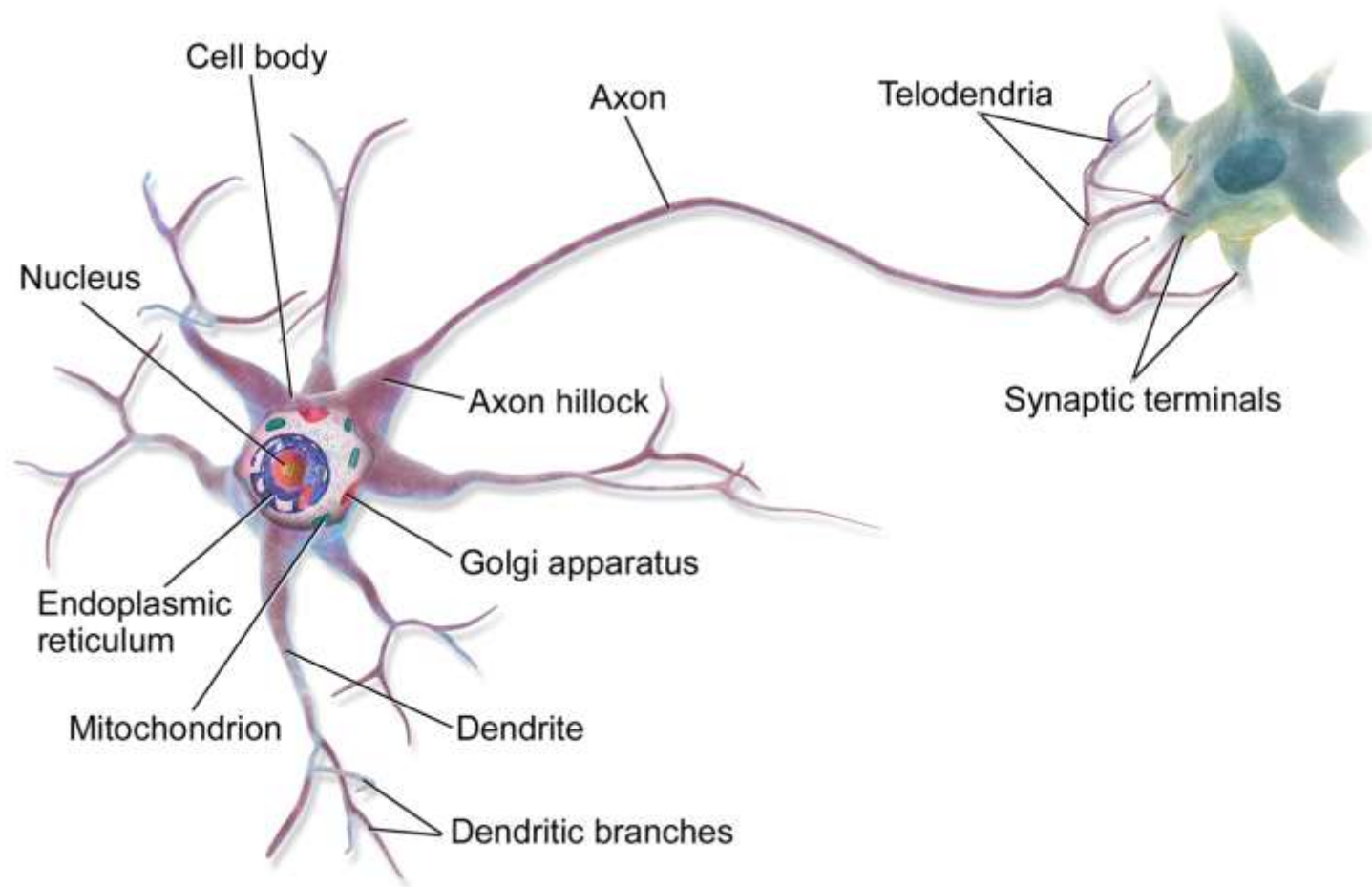
Lecture 1

Motivation

Deep Learning

- Deep learning is a branch of machine learning that uses neural networks with multiple layers to learn task-relevant representations from data.
- Each layer applies a learned transformation to the previous layer's output; deeper models can build increasingly complex features, although those features are not always human-interpretable.
- Deep neural networks have driven major advances in computer vision, speech, natural language processing, and scientific applications, but results depend on data, compute, model design, and evaluation.

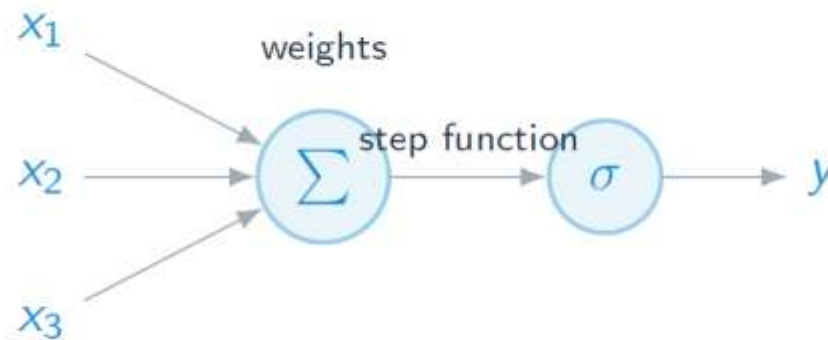
Biological Inspiration



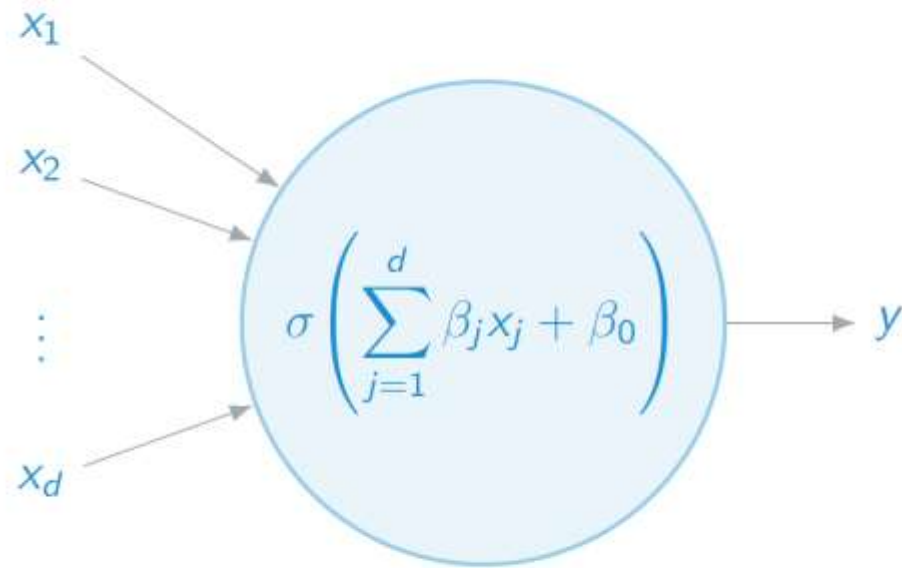
The artificial neuron

1958

- The perceptron was developed by Frank Rosenblatt, a psychologist and cognitive scientist, in 1958.
- It is considered one of the earliest models of artificial neural networks, inspired by the way biological neurons work



The artificial neuron



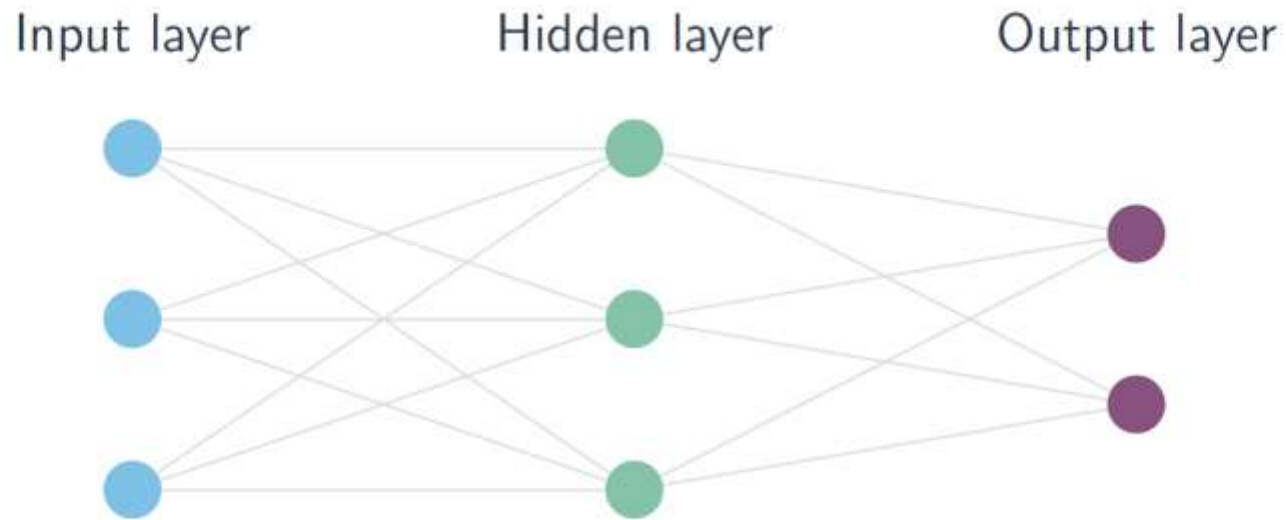
Simple perceptron

x_j is the j^{th} feature, β_j is its weight, and β_0 is the bias.

The bias shifts the decision boundary

$$\beta^T \mathbf{x} + \beta_0 = 0.$$

From Perceptron to layers



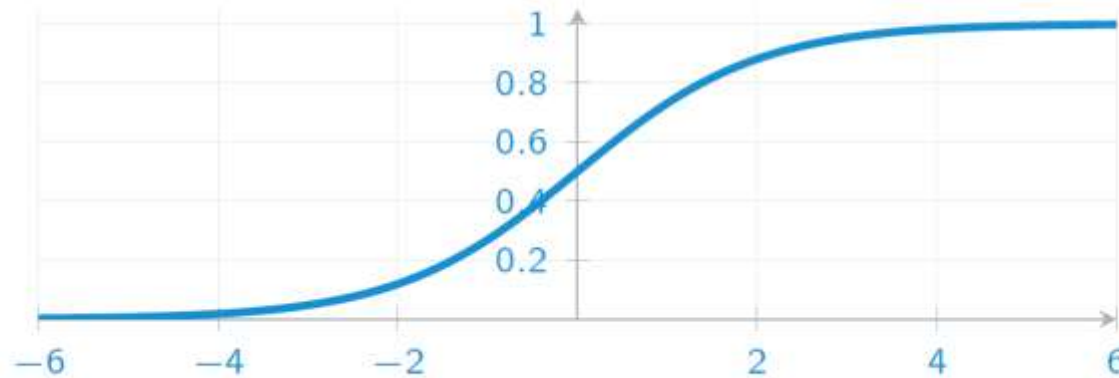
Neurons are connected by weighted edges; a layer transforms one representation into the next.

Activation functions

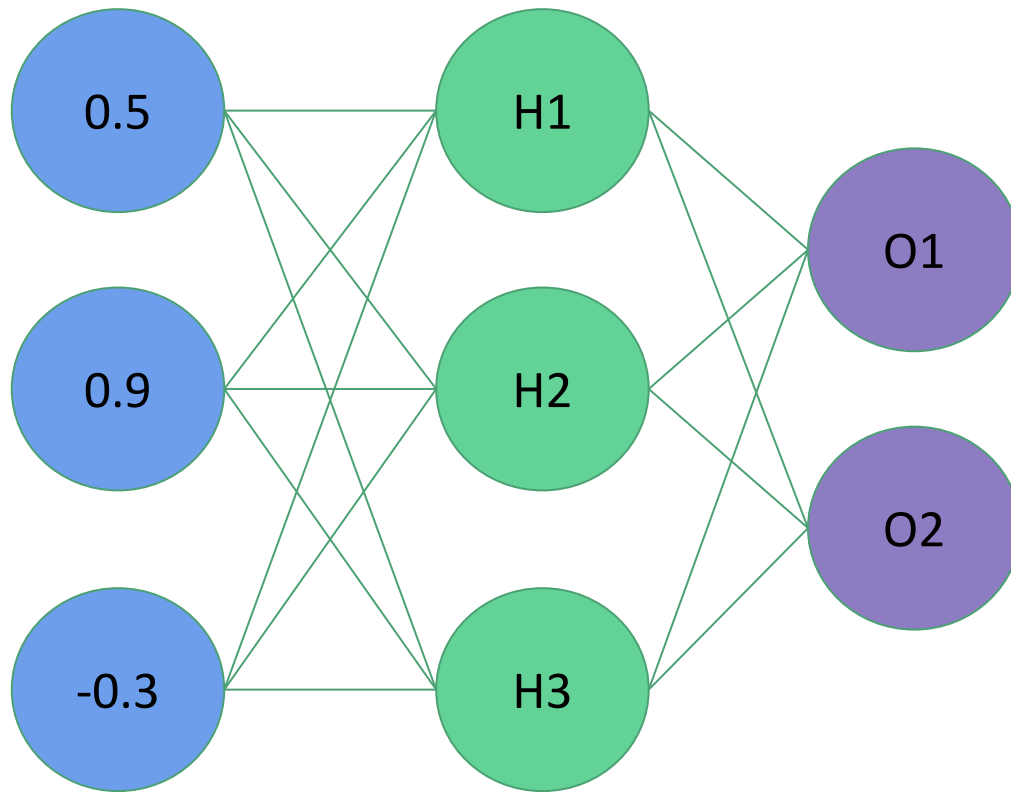
Activation functions are applied to the output at each neuron.

A classical example is the sigmoid:

$$\sigma(t) = \frac{1}{1 + e^{-t}}.$$



Neural Network



H1 Weights = (1.0, -2.0, 2.0)

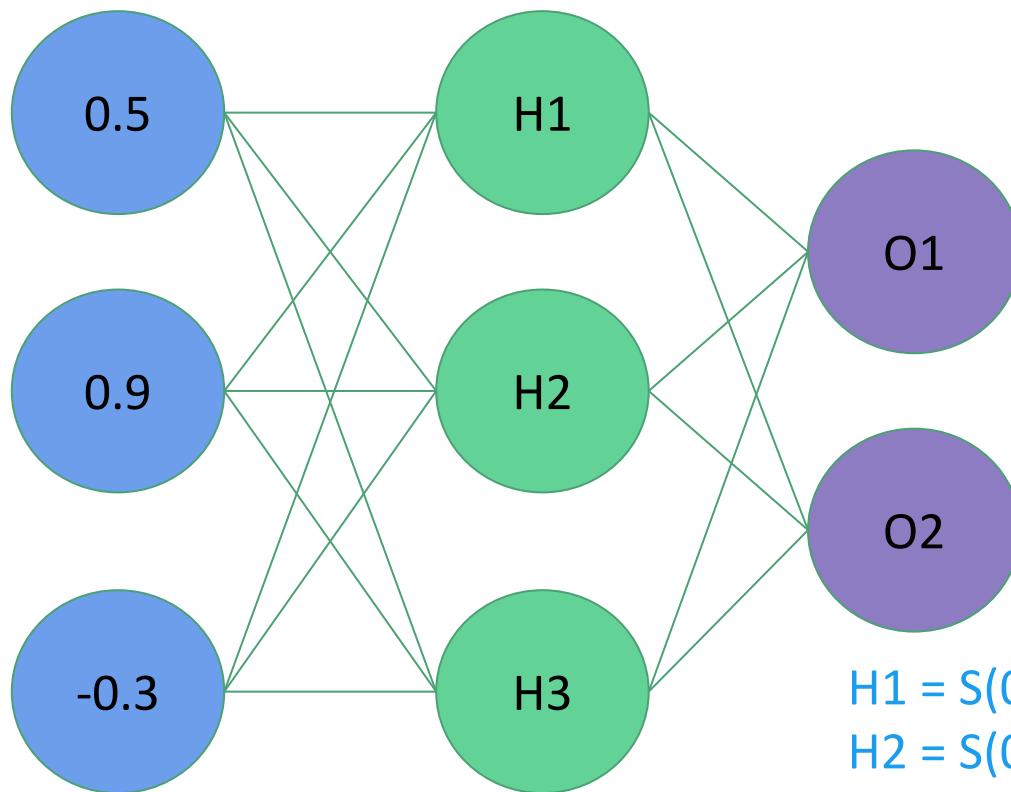
H2 Weights = (2.0, 1.0, -4.0)

H3 Weights = (1.0, -1.0, 0.0)

O1 Weights = (-3.0, 1.0, -3.0)

O2 Weights = (0.0, 1.0, 2.0)

Neural Network



H1 Weights = (1.0, -2.0, 2.0)

H2 Weights = (2.0, 1.0, -4.0)

H3 Weights = (1.0, -1.0, 0.0)

O1 Weights = (-3.0, 1.0, -3.0)

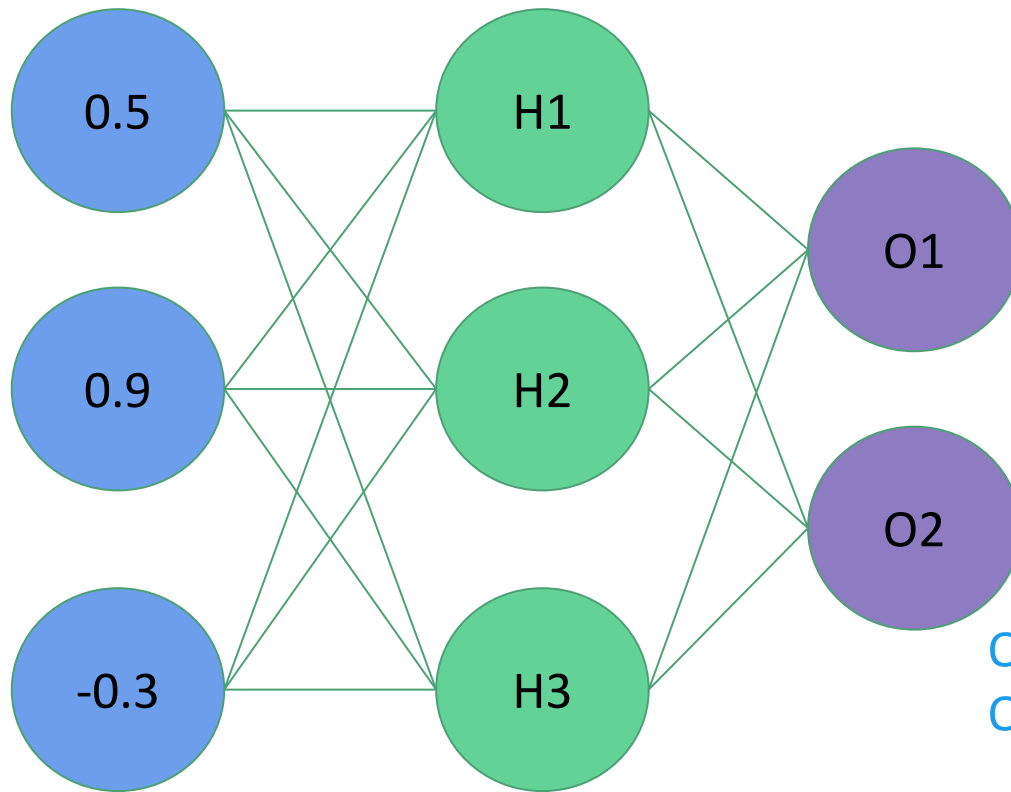
O2 Weights = (0.0, 1.0, 2.0)

$$H1 = S(0.5 * 1.0 + 0.9 * -2.0 + -0.3 * 2.0) = S(-1.9) = .13$$

$$H2 = S(0.5 * 2.0 + 0.9 * 1.0 + -0.3 * -4.0) = S(3.1) = .96$$

$$H3 = S(0.5 * 1.0 + 0.9 * -1.0 + -0.3 * 0.0) = S(-0.4) = .40$$

Neural Network



H1 Weights = (1.0, -2.0, 2.0)

H2 Weights = (2.0, 1.0, -4.0)

H3 Weights = (1.0, -1.0, 0.0)

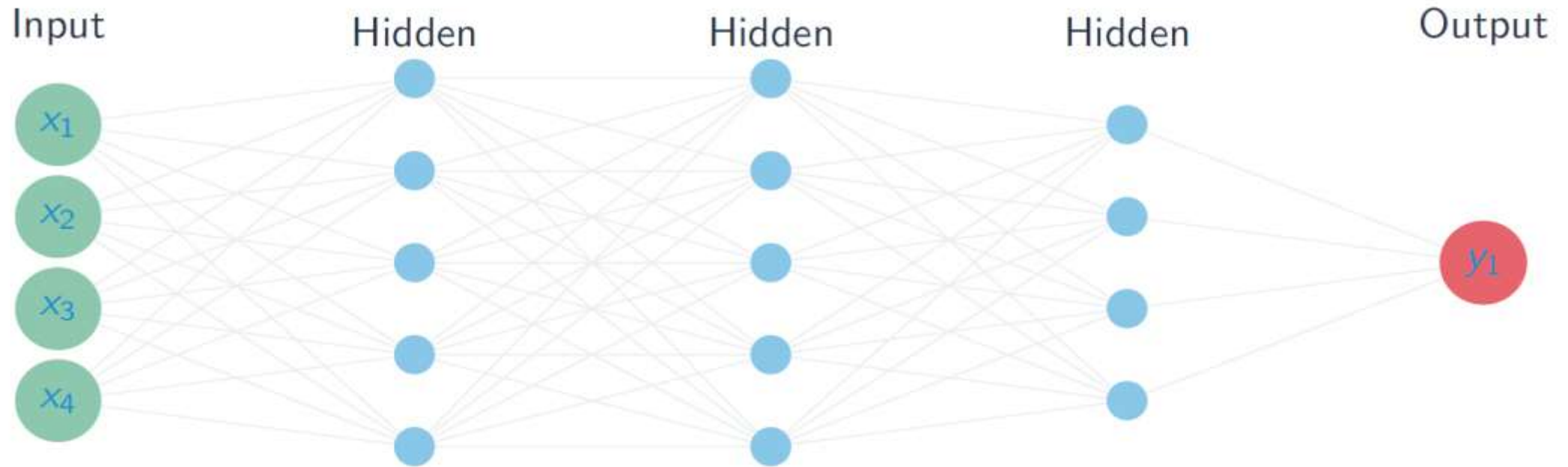
O1 Weights = (-3.0, 1.0, -3.0)

O2 Weights = (0.0, 1.0, 2.0)

$$O1 = S(.13 * -3.0 + .96 * 1.0 + .40 * -3.0) = S(-.63) = .35$$

$$O2 = S(.13 * 0.0 + .96 * 1.0 + .40 * 2.0) = S(1.76) = .85$$

Feedforward Deep Networks



A feedforward network composes layerwise transformations from input to output.



MILESTONES ACROSS THE FIELD — NOT JUST CHATBOTS

1 2006 — Deep belief nets
Revived effective deep-network training

2 2012 — AlexNet
Computer vision + GPU training

3 2015–17 — ResNet, AlphaGo, Transformer
Depth, reinforcement learning, attention

4 2020 — Scaling era
Large self-supervised foundation models

5 2021–22 — AlphaFold2 + diffusion
AI for science + generative vision

6 2024 — AlphaFold3 + multimodality
Broader scientific and multimodal systems

7 2025 — DeepSeek-R1
Reasoning through reinforcement learning

8 Feb 2026 — Search-interest spike
No single cause established

Source: user-provided Google Trends chart through February 2026. Values are normalized search interest; milestones indicate timing, not proven causation.

Deep learning progressed from learned features to reasoning

Selected milestones across vision, language, reinforcement learning, generative models, and science · 2006–2026

FOUNDATIONS

Representation learning and depth

- **2006 Deep belief networks**
Layer-wise pretraining revived effective training of deep architectures.
- **2012 AlexNet**
GPU-trained convolutional networks decisively won ImageNet.
- **2014–15 GANs and ResNet**
Adversarial generation and residual connections expanded what deep networks could learn.

ARCHITECTURES & TRAINING

Attention, reinforcement learning, and pretraining

- **2016–17 AlphaGo and Transformer**
Deep reinforcement learning mastered Go; self-attention replaced recurrence for sequence modeling.
- **2018–20 BERT, GPT-2, GPT-3**
Large pretrained models made transfer, prompting, and in-context learning central paradigms.
- **2020–22 AlphaFold2 and diffusion**
Deep learning reached protein structure prediction and high-quality image generation.

MULTIMODAL REASONING

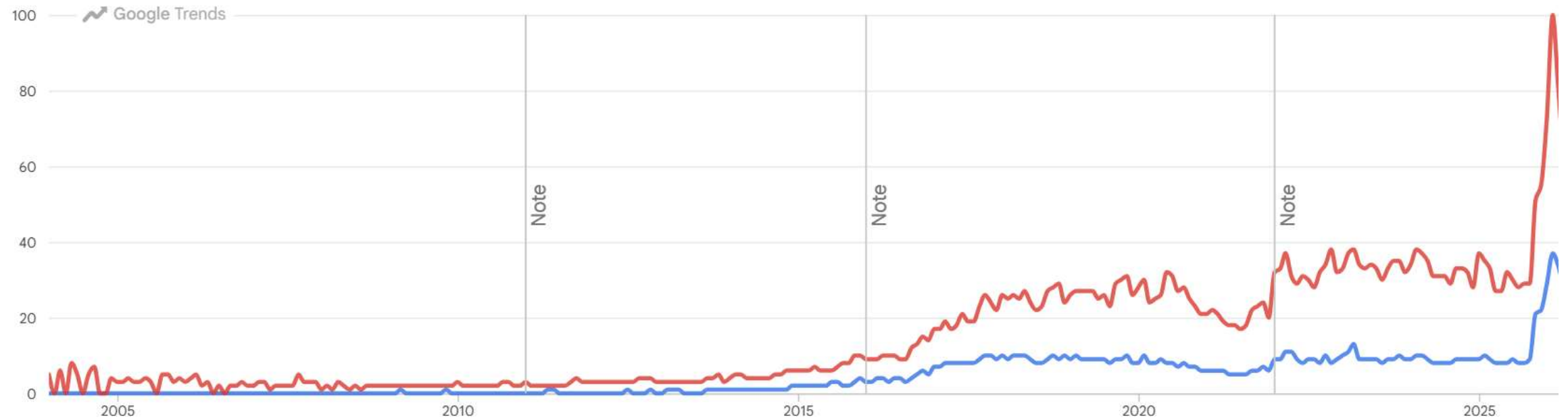
From conversational models to tool-using agents

- **2022–23 ChatGPT and GPT-4**
Instruction tuning broadened access; multimodal foundation models combined text and images.
- **2024–25 AlphaFold3 and reasoning**
Scientific prediction broadened, while test-time reasoning and DeepSeek-R1 highlighted reinforcement learning for reasoning.
- **2026 Agentic multimodal systems**
The frontier increasingly combines reasoning, tools, memory, and longer-running workflows.

The field's center of gravity has shifted from task-specific feature learning toward general models that reason across modalities and act through tools.

Sources: landmark papers and official releases from NeurIPS/CVPR, Google DeepMind, OpenAI, and DeepSeek. Updated September 2026.

● deep learning ● machine learning



Deep learning reaches nearly every data-rich domain

Major application areas to orient the landscape before we examine specific examples

LANGUAGE & DIGITAL

Understanding, generation, and personalization

● Text and language

Search, translation, summarization, question answering, coding assistants, and conversational agents.

● Speech and audio

Speech recognition, voice synthesis, music generation, transcription, and real-time translation.

● Recommendations and decisions

Personalized content, ranking, advertising, fraud detection, forecasting, and intelligent automation.

IMAGES, VIDEO & AUTONOMY

Perception, generation, and action

● Image understanding

Object detection, visual search, document understanding, quality inspection, and satellite analysis.

● Generative images and design

Image creation, editing, restoration, creative tools, simulation, and synthetic training data.

● Video, robotics, and autonomy

Video understanding and generation, driver assistance, navigation, robotics, and industrial control.

HEALTH & SCIENTIFIC DISCOVERY

Prediction and generation in complex domains

● Clinical care and medical imaging

Diagnostic support, radiology and pathology analysis, patient risk prediction, and workflow assistance.

● Biology and drug discovery

Protein and molecular modeling, target identification, molecule design, and clinical-trial optimization.

● Scientific and engineering

Materials, climate, weather, physics, and engineering models that accelerate simulation and discovery.

The key shift is convergence: modern systems increasingly combine text, images, audio, video, scientific data, and tool use in one workflow.

Application map for orientation; categories overlap as multimodal models connect digital, physical, clinical, and scientific workflows.

An Early Book Created with Generative AI

Alice and Sparkle Paperback – Jan. 14 2023

by Ammaar Reshi (Author), Chat GPT (Author), Mid Journey (Illustrator)

3.1 ★★★★★ 30 ratings

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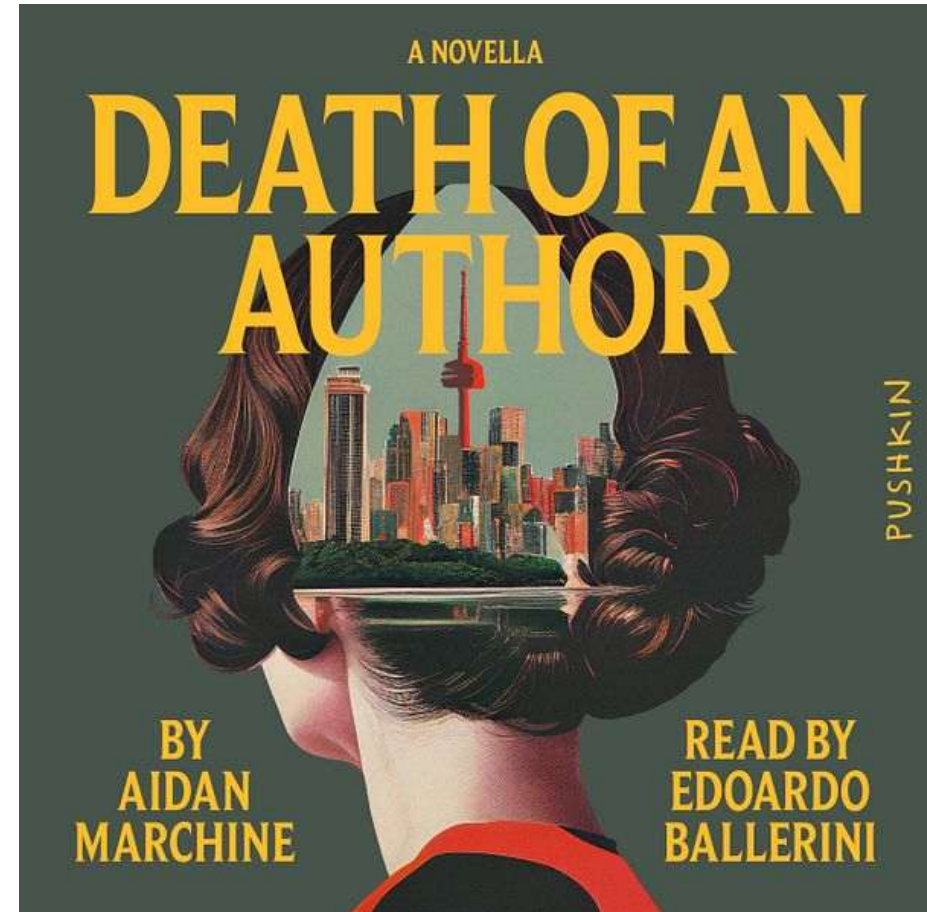
The children's book covered in The Washington Post, TIME Magazine, Le Figaro, NBC News, and media worldwide.

This is a story about a young girl named Alice who discovers the magic of artificial intelligence. She creates her own AI, named Sparkle, and together they go on adventures and use their combined knowledge to make the world a better place. The story explores the incredible abilities of AI and the importance of using them for good. It is a tale of friendship and exploration, filled with magic and wonder.



A Novella Written Through Human–AI Collaboration

- Published in 2023, *Death of an Author* was written by Stephen Marche, using the pen name Aidan Marchine, in collaboration with three AI systems.



By 2026, AI-Generated Books Became a Market Issue

- **The 2023 figure was only an early snapshot:** More than 200 Kindle titles explicitly listed ChatGPT as an author or co-author; most AI use was not publicly identified.
- **A 2026 study examined 14,419 self-published genre-fiction books** sold on Amazon from 2023 through June 2026.
- **AI text had reached commercial scale:** Books with substantial AI text were 20% of the sample but generated 12.1% of sales; together, light and substantial AI books occupied about 40% of sampled Top-25 positions by 2026.

• Sources: Amazon KDP; U.S. Copyright Office; Chakrabarty et al., arXiv:2607.20349 (July 2026).

Sentence Styles by Different Writers

- **Sentence Example:** "The sun set over the distant hills, painting the sky with hues of orange and pink."
- **William Shakespeare Style:** "The fiery orb did sink beyond the far-off hills, brushing the heavens with a palette of burnished gold and blushing rose."
- **Gabriel García Márquez Style:** "The sun dissolved behind the hills, spilling its final breaths of life in a riot of oranges and pinks, as if the sky itself bled into the horizon."
- **Jorge Luis Borges Style:** "The sun, like an ancient god retreating to the depths of its labyrinthine domain, disappeared behind the distant hills, casting the sky in enigmatic shades of orange and pink—a fleeting moment in the infinite repetition of time."

Generative AI Beyond Text and Images

- **Text-to-Music and Text-to-3D Generation:**
- **Jukebox:** A model that generates music, including lyrics, in various genres and styles. It can mimic the style of famous artists or create entirely new compositions.
- **DreamFusion and Point-E:** Models for generating 3D models from text descriptions, enhancing fields like gaming, augmented reality (AR), and virtual reality (VR).

Genie 3 Turns a Prompt into a Navigable World

- **From text to environment:** A prompt generates a dynamic, interactive world.
- **Real-time interaction:** Users can move through and change the environment as it is generated.
- **Why it matters:** World models can provide open-ended environments for training and evaluating agents.
- Source: Google DeepMind, August 2025.



The background of the slide is a dark, semi-transparent overlay on a video frame. The video frame shows a reflection in a spherical mirror, likely from the movie 'E.T. the Extra-Terrestrial'. The reflection depicts a child's playroom with a wooden floor, a small wooden toy house in the foreground, and shelves filled with toys and books in the background. The text 'Project Genie' is centered in the upper half, with a small 'EXPERIMENT' button to its right. Below this, the words 'Image Upload' are written in a large, bold, white font.

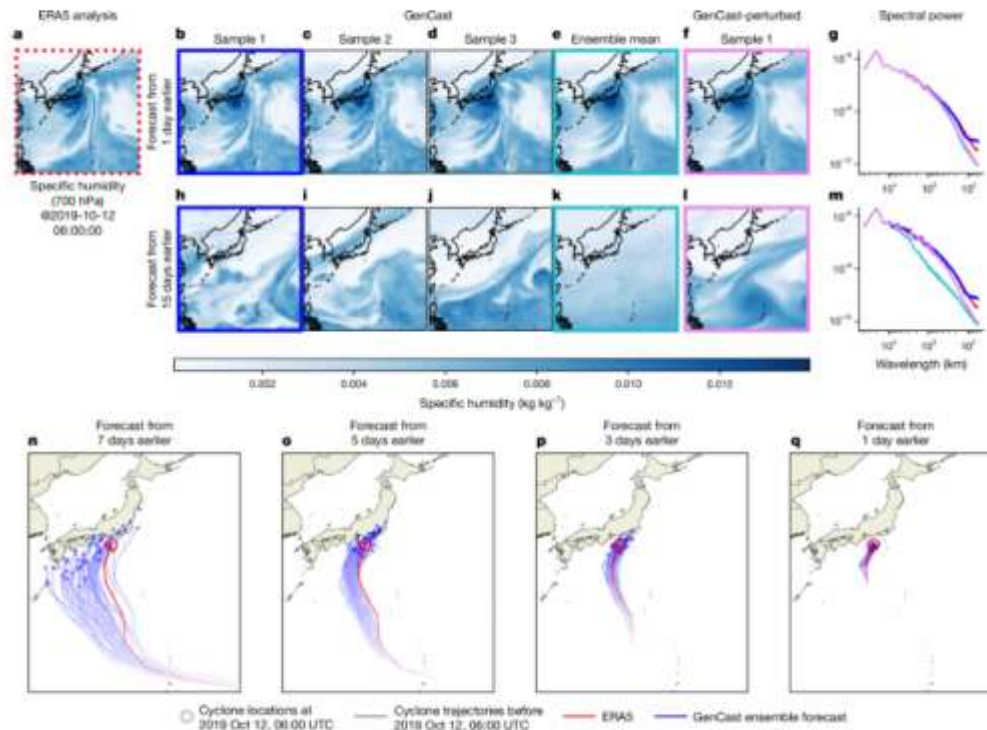
Project Genie EXPERIMENT

Image Upload

AI for Scientific Discovery

- **AI in Climate Science:**
 - Deep learning models are used to predict climate patterns, simulate climate models, and forecast extreme weather events. Models like **DeepMind's deep-learning weather forecast systems** and **ClimateBERT** provide more accurate predictions of weather phenomena, aiding in disaster preparedness and resource management.

Scientific discovery: global weather



GenCast

15 days
global ensemble forecast

8 min
generation time

Better than the operational ENS
system on **97.2% of 1,320**
evaluated targets.

AI for Scientific Discovery

- **AI in Physics and Material Science:**
 - AI models help in discovering new materials by predicting material properties and simulating chemical reactions. Tools like **MatBERT** and **SchNet** are used for molecule property prediction and materials discovery, advancing fields like superconductors and battery technology.

Scientific discovery: new materials

GNoME

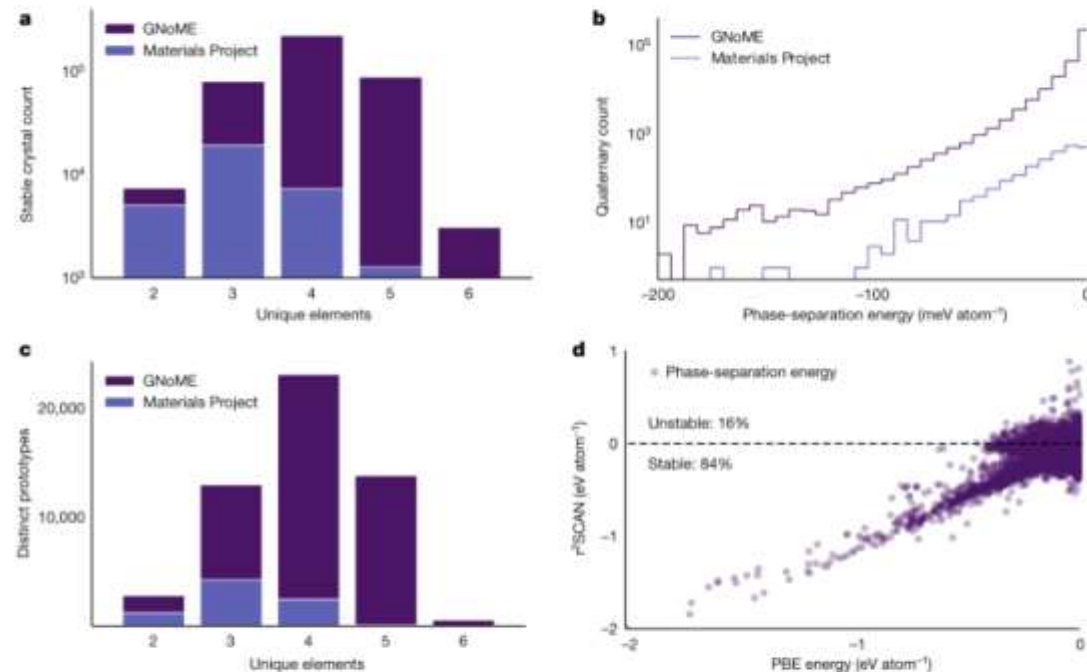
2.2 M

stable structures

381 k

new structures on the updated
convex hull

Deep learning turns an enormous
combinatorial search space into a
ranked set of candidates for
computation and experiment.



Source: Google DeepMind, Nature, 2023.

GNoME Expands the Search for New Materials

- **Scale:** Predicted 2.2 million new crystal structures.
- **High-priority candidates:** About 380,000 were predicted to be especially stable.
- **How it works:** A graph neural network proposes and ranks structures; density-functional calculations verify stability.
- **What changes:** Researchers can focus experiments on a much smaller, higher-value set of candidates.
- **Possible applications:** Batteries, electronics, superconductors, and other material technologies.
- **Important limit:** Predicted stability does not guarantee that a material can be synthesized or will be useful.
- Source: Google DeepMind, Nature, 2023.

AI for Scientific Discovery

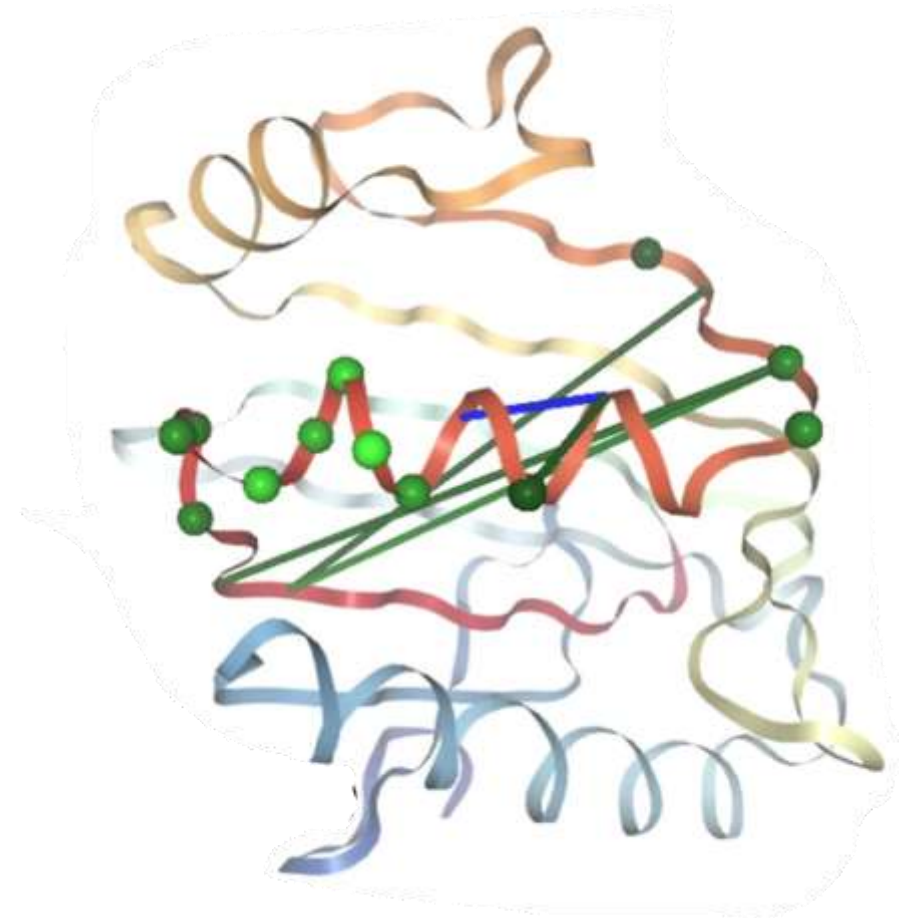
- **AI-Driven Research and Simulations:**
 - Deep learning accelerates scientific research by enabling faster and more efficient simulations of complex physical systems. AI is being leveraged to study black holes, quantum mechanics, and particle physics, pushing the boundaries of knowledge in fundamental sciences.

AI in Healthcare Diagnostics

- **AI-Assisted Radiology:**
 - **CheXNet:** A deep learning model that detects pneumonia from chest X-rays, outperforming radiologists in some cases. It demonstrates the potential of AI to assist doctors in diagnostic accuracy and speed.
 - **AI in MRI/CT Scans:** Models that identify brain tumors, cardiovascular issues, and other anomalies with high precision.

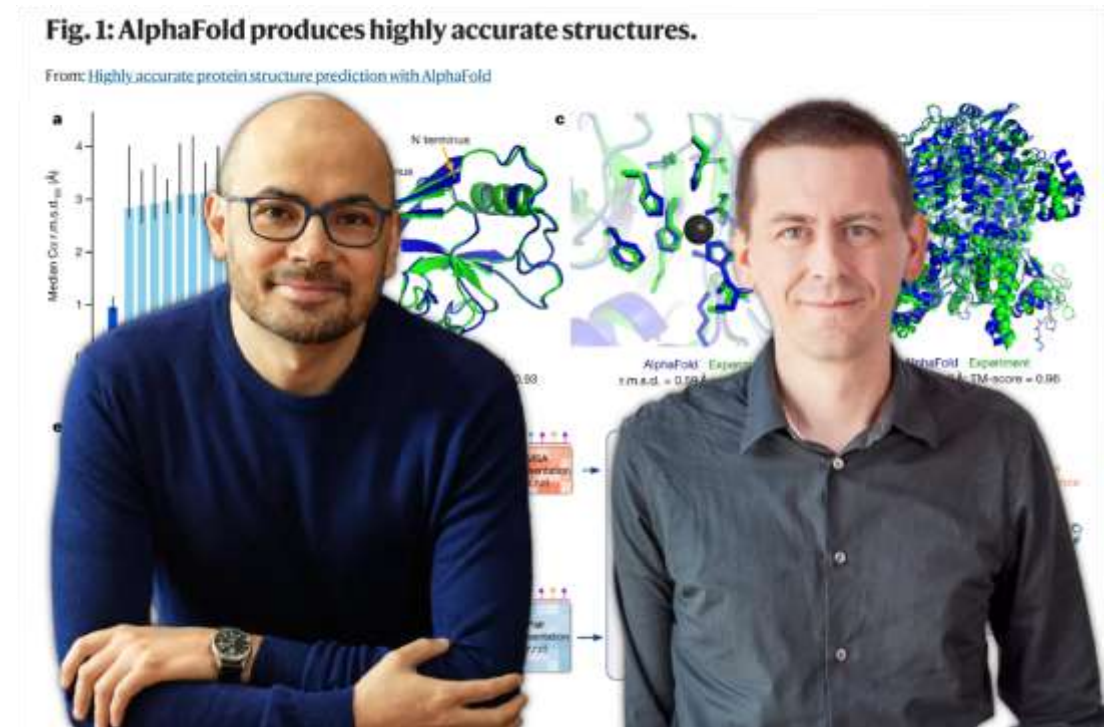
AlphaFold 2 Predicts Protein Structure

- **The problem:** Protein shape determines function, but experiments are slow and costly.
- **AlphaFold 2:** Predicts 3D structure from an amino-acid sequence with near-experimental accuracy.
- **Deep learning:** Uses evolutionary patterns and attention to model distant interactions.
- **Health-care impact:** Speeds research on disease proteins and drug targets.
- Source: Jumper et al., Nature, 2021.



Nobel Prize in Chemistry (2024): Computer Scientists, Not Chemists

- AlphaFold2 earned Demis Hassabis and John Jumper the 2024 Nobel Prize in Chemistry for transforming protein-structure prediction.



AlphaFold 3 Models Molecular Complexes

- **Released in May 2024:** AlphaFold 3 expands prediction beyond individual proteins.
- **What it models:** Complexes containing proteins, DNA, RNA, small molecules, ions, and modified residues.
- **Key change:** A diffusion-based module generates atomic coordinates for the interacting system.
- **Why it matters:** Better models of drug–target and DNA–protein interactions may support biological research and drug discovery.
- **Limit:** Predictions are mainly static models and still require experimental validation.
- Source: Abramson et al., Nature, 2024.

IMO 2025: AI Reached Gold-Medal-Level Reasoning

- **Score:** 35 of 42 points—the gold-medal threshold.
- **Problems solved:** Five of six problems received full credit.
- **Format:** Gemini Deep Think produced complete proofs in natural language.
- **Conditions:** The system worked within the official competition time limit.
- Source: Google DeepMind, July 2025.

The Jacobian Conjecture

- **The idea, proposed in 1939:** A well-behaved polynomial transformation should be reversible.
- **In plain language:** If a transformation does not locally crush or fold space, can two different inputs ever produce the same output?
- **What changed in July 2026:** An AI-assisted search produced a three-variable example in which three different inputs give the same output.
- **Current status:** The conjecture is false in three or more dimensions. The two-variable case remains open.
- Source: Shuhong Gao, arXiv:2608.00222.

AI Tackles a 92-Year-Old Fluid Problem

- Open since 1934: The Navier–Stokes regularity problem remained unresolved for 92 years.
- In simple language: If a fluid starts smooth, can its motion suddenly become infinitely violent at one point—or must it always remain well-behaved?
- Why it matters: Navier–Stokes equations underpin models of aircraft, turbines, engines, pipelines, weather, oceans, and blood flow.
- Possible impact: A proof could improve understanding of extreme flows and the reliability of numerical simulations, although it would not provide a universal fluid solver.
- September 8, 2026: OpenAI released an AI-generated candidate proof together with a machine-checked Lean formalization.
- Current status: Independent mathematical validation and broad acceptance are still pending.

Forced Navier–Stokes blow-up theorem

For every positive viscosity $\nu > 0$, one can construct a three-dimensional incompressible flow that starts completely at rest,

$$u(x, 0) = 0$$

A completely smooth external force is then applied. In finite time, in this construction, at $t = 1$, the velocity in a very small region grows without bound:

$$\|u(t)\|_{\infty} \rightarrow \infty \quad \text{as} \quad t \rightarrow 1$$

At the same time, the total energy of the fluid remains bounded:

$$\|u(t)\|_2 < C$$

The Controversy: Discovery or Data Influence?

- **September 8, 2026:** OpenAI announced that about 10,000 coordinated AI agents produced a candidate finite-time blowup proof for a forced 3D Navier–Stokes equation in 88 hours, with a Lean formalization.
- **One day earlier:** Tristan Buckmaster and Levent Alpöge released related blowup results for other fluid equations after roughly a year of work.
- **The overlap:** Both efforts used an unusual “smooth forcing” strategy—an external mathematical input designed to drive the fluid toward a singularity.
- **Buckmaster’s concern:** He said his team had discussed private work through Codex and asked whether those interactions could have influenced model training.
- **Why it became controversial:** A correct proof would not by itself reveal whether the key idea was independently generated, learned from public research, or influenced by user interactions.

A Similar Provenance Debate Had Already Emerged

- **The earlier announcement:** OpenAI's "Astra" project claimed progress on ten open mathematical problems.
- **Andreas Thom's concern:** He argued that one proof drew directly on a 2019 Kun–Thom paper and noted that he had spent months discussing possible extensions with ChatGPT.
- **Other researchers' concern:** Steven Miller and others said prior work was ignored or understated; a claim that the area had seen "no progress for a decade" was later revised.
- **OpenAI's position:** The systems did not retrieve private drafts or user data while producing the proofs, and the resulting arguments differed from earlier work.
- **The unresolved issue:** That does not fully answer whether de-identified product usage may previously have contributed to model improvement or training.

What remains genuinely hard?

RELIABILITY

Know when the model is wrong

CAUSALITY

Reason beyond correlations

CONTINUAL LEARNING

Learn without forgetting

EMBODIMENT

Act safely in the physical world

EVALUATION

Measure capability and risk

Impressive demonstrations are not the same as a complete theory – or a dependable system.

A compact history of deep learning



The ideas are old; the inflection came from **data, compute, optimization, and scalable architectures.**

Machine Learning

Supervised learning

Learn from labeled input–output examples.

- Classification
- Regression

Unsupervised learning

Find structure without externally supplied targets.

- Clustering
- Dimensionality reduction

Reinforcement learning

Learn actions from rewards through interaction with an environment.

Classification

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n, \quad x_i \in \mathbb{R}^d, \quad y_i \in \mathcal{Y}, \quad |\mathcal{Y}| < \infty.$$

Find a function

$$f : \mathbb{R}^d \rightarrow \mathcal{Y}$$

so that for a new input x , the prediction is $f(x)$.

Regression

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n, \quad x_i \in \mathbb{R}^d, \quad y_i \in \mathbb{R}.$$

Find a function

$$f : \mathbb{R}^d \rightarrow \mathbb{R}$$

so that for a new input x , the predicted response is $f(x)$.

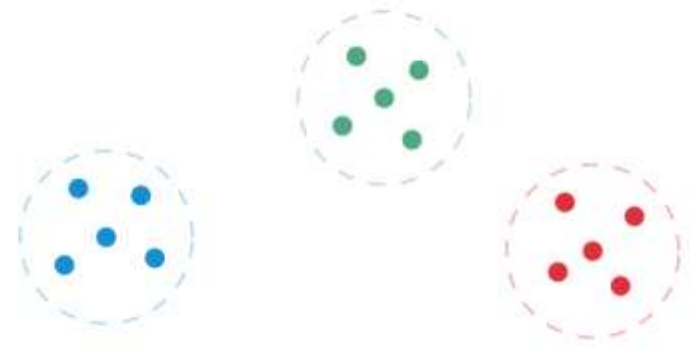
Clustering

$$\mathcal{D} = \{x_i\}_{i=1}^n, \quad x_i \in \mathbb{R}^d.$$

Find an assignment

$$f : \mathbb{R}^d \rightarrow \{1, \dots, K\}$$

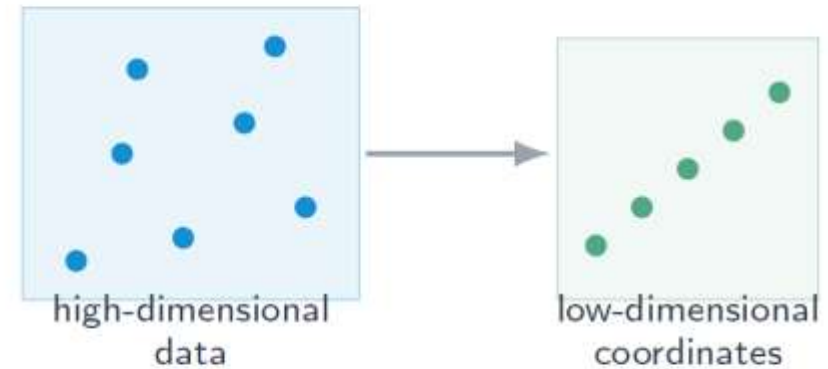
such that similar observations receive the same cluster label.



Dimensionality reduction

Represent high-dimensional data with fewer coordinates while preserving useful structure.

- Examples: PCA, manifold learning, autoencoders
- Uses: visualization, compression, denoising, features



Beyond the classical four

Adapting with limited examples

- Zero-shot learning
- One-shot and few-shot learning
- In-context learning

Learning across tasks and time

- Transfer learning
- Multi-task learning
- Meta-learning
- Continual learning

These regimes do not replace the classical task types.

They describe how a model transfers, adapts, or continues learning across tasks.

What is Machine Learning?

- Machine Learning is just glorified "curve fitting"?

General Procedure

1. Model (Hypothesis class)

$$f(\mathbf{x}) \in \mathcal{F} \text{ (Hypothesis class)}$$

2. Score Criterion

$$\text{Population: } S(f) = E_{\mathbf{x}, y} L(y, f(\mathbf{x}))$$

$$\text{Sample: } \frac{1}{n} \sum_{i=1}^n L(y_i, f(\mathbf{x}_i))$$

3. Search Strategy

$$\hat{f} = \arg \min_{f \in \mathcal{F}} S(f)$$

Model

(Hypothesis class)

- Linear (Perceptron, SVM, Logistic regression)
- Directed graphical model
- Undirected graphical model
- Matrices

Score criterion

Cost for error: $L(y, F)$

$$L(y, F) = |y - F|, (y - F)^2, \quad y \in \mathcal{R}$$

$$y \in \{-1, 1\} :$$

$$L(y, F) = \log(1 + e^{-yF}) \quad \text{logistic reg.}$$

$$L(y, F) = \max(0, 1 - yF) \quad \text{SVM}$$

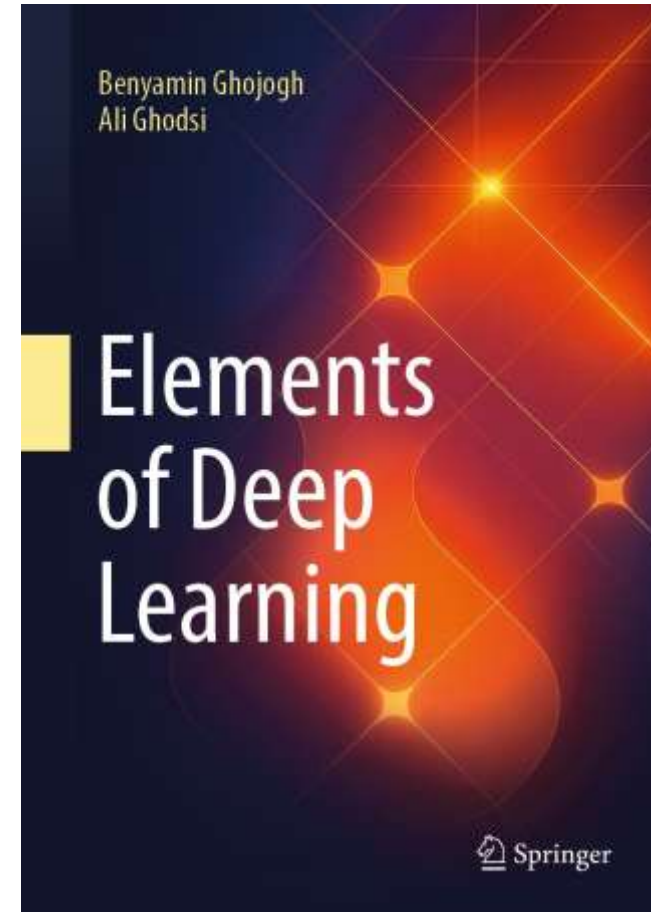
Many many more

Deep Structure as a Hypothesis class

- Feedforward Deep Networks
- Convolutional Networks
- Recurrent neural networks (RNN)
- Restricted Boltzmann Machines (RBM)

Course Texts

- **Required textbook**
Benyamin Ghojogh and Ali Ghodsi, Elements of Deep Learning, Springer, 2026.



Course Platforms

- **Course website**
Materials, readings, schedules, instructions, and resources.
- Go to aghodsib.github.io → Teaching → Deep Learning.
- **Piazza**
Course communication, questions, and private feedback. Access it through the course-website widget.
- **Learn**
Dropboxes and grades.
- Students are responsible for checking all official platforms regularly.

Communication Policy

- Use Piazza for all communication with the instructor and teaching assistant.
- **Public posts**
Questions and answers that are useful to the whole class.
- **Private posts**
Personal matters and private feedback.
- Do not email the instructional team directly about course matters.

Assessment and Evaluation

- **Assignments — 45%**
Three individual assignments, each worth 15%.
- **Paper presentation — 10%**
Groups of up to four students; every member must participate and answer questions.
- **Group final project — 45%**
Groups of up to four students submit one final report.
- **Total — 100%**

Presentations and Projects

- **Paper presentation**

Up to four students; each member must participate and answer questions.

- Eligible venues: NeurIPS, ICML, ICLR, AISTATS, AAAI, IJCAI, UAI, CVPR, ICCV, ECCV, ACL, EMNLP, NAACL, JMLR, TMLR, IEEE TPAMI, or Artificial Intelligence.

- **Final project**

Groups of up to four submit one report. Work must be new for STAT 940 and may not reuse thesis work or work submitted for another course.

- Presentation and project groups may be the same or different.
- Students may be asked to explain submitted work orally or complete a short individual quiz to confirm authorship and understanding.

Tentative Assessment Dates

- **Assignment 1**
Released October 5 · Due October 19 at 11:59 p.m.
- **Assignment 2**
Released November 2 · Due November 16 at 11:59 p.m.
- **Assignment 3**
Released November 23 · Due December 7 at 11:59 p.m.
- **Final project**
Due December 21 at 11:59 p.m.; detailed requirements to be announced.
- All dates are tentative; changes will be announced.

Course Policies

- **Late written work**
Within 24 hours: 10% penalty on the earned mark. After 24 hours: grade of zero.
- **Generative AI**
May support learning unless an assessment states otherwise.
- Submitted work must reflect the student's own authorship and understanding; substantive use must be disclosed.
- Students must verify all work, citations, code, and results.
- Generative AI is prohibited during oral or quiz-based verification. Unauthorized or undisclosed use may constitute an offence under Policy 71.

Tentative course plan and textbook alignment

Tentative topic	Textbook alignment
Foundations, perceptron, and feedforward neural networks	Ch. 1, Secs. 2-3 and 5; Ch. 2, Secs. 2.2-2.9 and 4
Backpropagation and optimization	Ch. 3, Secs. 2-5
Model selection, regularization, and generalization	Ch. 5, Secs. 4-11; Ch. 6, Secs. 2-3
Convolutional neural networks	Ch. 4, Secs. 2-3 and 5.2-5.7
Recurrent neural networks and sequence-to-sequence models	Ch. 8, Secs. 2-5; Ch. 9, Secs. 2.1-2.3
Attention, Transformers, and Performer	Ch. 9, Secs. 2.3-4.2
Large language models and alignment	Ch. 11, Secs. 2-4 and 7-9
Variational models, moment matching, and GANs	Ch. 12, Secs. 2-3; Ch. 13, Secs. 2-5; Ch. 14, Secs. 2-3
Diffusion models	Ch. 15, Secs. 2-4
Graph neural networks	Ch. 17, Secs. 2-7
Deep reinforcement learning, if time permits	Ch. 18, Secs. 2-9
Presentations and selected emerging topics	Assigned papers; no fixed textbook section